

**Foundation Programme**  
**Universiti Teknologi Malaysia**  
**Session 2020/2021 Semester 3**

Test 1 (15%)

Time: 1 hour 15 minutes

**FPSM 0024 Calculus**

*Answer all questions. Show all your work clearly.*

1. Given function  $f$  such that

$$f(x) = \begin{cases} 6x^2 + p, & x < 2 \\ q, & x = 2 \\ px^3 + 3, & x > 2 \end{cases}$$

- (a) Find the value of  $p$  such that  $\lim_{x \rightarrow 2} f(x)$  exists.

(3 marks)

- (b) Hence, find the value of  $q$  such that  $f(x)$  is continuous at  $x = 2$ .

(3 marks)

2. By using the first principle, find the derivative of  $f(x) = \frac{1}{\sqrt{x+3}}$ .

(4 marks)

3. Use appropriate rules to find  $\frac{dy}{dx}$  for  $y = e^{-2x} \cot(\sqrt{3x^2 + 4})$ .

(4 marks)

4. If  $y = e^x \sin(3 - 2x)$ , show that

$$\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + 5y = 0.$$

(5 marks)

5. Given  $y^3 = \ln(xy^6)$  for  $x > 0$  and  $y > 0$ . Find  $\frac{dy}{dx}$  in terms of  $x$  and  $y$ .

Hence, find  $\frac{dy}{dx}$  when  $x = e$  and  $y = 1$ .

(4 marks)

*Continue next page*

6. A water tank in the shape of right-circular cone with radius of 9 m (metre) and height of 12 m is used to contain a volume of water as shown in Figure 1 below. Given that  $h$  is the depth of the water in the tank and  $r$  is the radius of the top surface of the water. When there is a leaking at the base of the tank, the volume of the water left inside is modelled by

$$V = \frac{1}{3}\pi(972 - 12r^2 + r^2h).$$

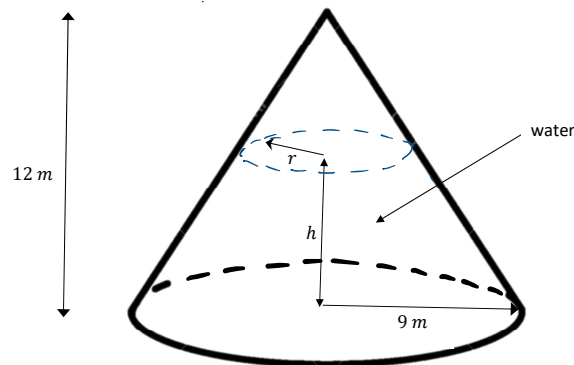


Figure 1: Figure for Question 6

- (a) Show that  $h = 12 - \frac{4}{3}r$ .

(2 marks)

- (b) Find  $\frac{dV}{dr}$  in terms of  $r$  only.

(2 marks)

- (c) Find  $\frac{dr}{dt}$  if  $r = 4$  m and the volume of water inside the tank is decreasing at the rate of  $3 \text{ m}^3/\text{hour}$ .

$$\frac{dV}{dt} = -3$$

(3 marks)

$$\begin{aligned} \frac{dr}{dt} &= \frac{dr}{dV} \times \frac{dV}{dt} \\ &= \frac{1}{16\pi r - \frac{8}{9}\pi r^2} \times -3 = -0.0192 \text{ ms}^{-1} \end{aligned}$$

END OF QUESTIONS

$$\begin{aligned} \frac{dV}{dr} &= 0 + 8\pi r + \frac{2\pi(12 - \frac{4}{3}r)r}{3} \\ &= 8\pi r + 8\pi r - \frac{8}{9}\pi r^2 \\ &= 16\pi r - \frac{8}{9}\pi r^2 \end{aligned}$$

## LIST OF FORMULA

| Trigonometric                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\cos^2 x + \sin^2 x = 1$ $1 + \tan^2 x = \sec^2 x$ $\cot^2 x + 1 = \operatorname{cosec}^2 x$ $\sin 2x = 2 \sin x \cos x$ $\cos 2x = \cos^2 x - \sin^2 x$ $= 2 \cos^2 x - 1$ $= 1 - 2 \sin^2 x$ $\tan 2x = \frac{2 \tan x}{1 - \tan^2 x}$ $\sin(x \pm y) = \sin x \cos y \pm \cos x \sin y$ $\cos(x \pm y) = \cos x \cos y \mp \sin x \sin y$ $\tan(x \pm y) = \frac{\tan x \pm \tan y}{1 \mp \tan x \tan y}$ $2 \sin x \cos y = \sin(x + y) + \sin(x - y)$ $2 \sin x \sin y = -\cos(x + y) + \cos(x - y)$ $2 \cos x \cos y = \cos(x + y) + \cos(x - y)$ |
| Logarithm                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| $a^x = e^{x \ln a}$ $\log_a x = \frac{\log_b x}{\log_b a}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

| Differentiations                                                        |
|-------------------------------------------------------------------------|
| $\frac{d}{dx}[k] = 0, \quad k \text{ constant}$                         |
| $\frac{d}{dx}[x^n] = nx^{n-1}$                                          |
| $\frac{d}{dx}[e^x] = e^x$                                               |
| $\frac{d}{dx}[\ln  x ] = \frac{1}{x}$                                   |
| $\frac{d}{dx}[\cos x] = -\sin x$                                        |
| $\frac{d}{dx}[\sin x] = \cos x$                                         |
| $\frac{d}{dx}[\tan x] = \sec^2 x$                                       |
| $\frac{d}{dx}[\cot x] = -\operatorname{cosec}^2 x$                      |
| $\frac{d}{dx}[\sec x] = \sec x \tan x$                                  |
| $\frac{d}{dx}[\operatorname{cosec} x] = -\operatorname{cosec} x \cot x$ |

$$1) f(x) = 6x^2 + p, \quad x < 2$$

$$= q, \quad x = 2$$

$$= px^3 + 3, \quad x > 2$$

$$(a) \lim_{x \rightarrow 2} f(x) \text{ is exist}$$

$$\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^+} f(x)$$

$$6x^2 + p = px^3 + 3$$

$$6(4) + p = 8p + 3$$

$$7p = 21$$

$$p = 3$$

$$(b) \lim_{x \rightarrow 2} f(x)$$

$$f(2) = q$$

$$\lim_{x \rightarrow 2^-} f(x) = 6(2^2) + 3$$

$$= 27$$

$$\lim_{x \rightarrow 2^+} f(x) = 3(2^3) + 3$$

$$= 27$$

$$\lim_{x \rightarrow 2} f(x) = f(2) = q = 27$$

$$2) f(x) = \frac{1}{\sqrt{x+3}}$$

$$f(x+h) = \lim_{h \rightarrow 0} \frac{\frac{1}{\sqrt{x+h+3}} - \frac{1}{\sqrt{x+3}}}{h}$$

$$= \frac{\frac{\sqrt{x+3} - \sqrt{x+h+3}}{\sqrt{x+h+3}\sqrt{x+3}} \times \frac{\sqrt{x+3} + \sqrt{x+h+3}}{\sqrt{x+3} + \sqrt{x+h+3}}}{h}$$

$$= \frac{x+3 - x-h-3}{h(\sqrt{x+h+3}\sqrt{x+3} + \sqrt{x+3}\sqrt{x+h+3})}$$

$$= \frac{-h}{h(2\sqrt{x+3}\sqrt{x+h+3})}, \quad h \neq 0$$

$$= \frac{-1}{2\sqrt{x+3}\sqrt{x+3}}$$

$$3) y = e^{-2x} \cot(\sqrt{3x^2+4})$$

$$u = e^{-2x} \quad u' = -2e^{-2x}$$

$$v = \cot(\sqrt{3x^2+4}) \quad v' = -\operatorname{cosec}^2(\sqrt{3x^2+4}) \times \frac{3x}{\sqrt{3x^2+4}}$$

$$\frac{dy}{dx} = u'v + uv'$$

$$= \left[ e^{-2x} \left( -\frac{3x \operatorname{cosec}^2(\sqrt{3x^2+4})}{\sqrt{3x^2+4}} \right) \right] + \left[ -2e^{-2x} \cot(\sqrt{3x^2+4}) \right]$$

$$= \frac{-3xe^{-2x} \operatorname{cosec}^2(\sqrt{3x^2+4})}{\sqrt{3x^2+4}} + \frac{-2e^{-2x} \cos \sqrt{3x^2+4}}{\sin \sqrt{3x^2+4}}$$

$$= \frac{-3x}{e^{2x} \sqrt{3x^2+4}} + \frac{-2 \cos \sqrt{3x^2+4}}{e^{2x} \sin \sqrt{3x^2+4}} = \frac{-3x}{e^{2x} \sqrt{3x^2+4} \sin^2(\sqrt{3x^2+4})} - \frac{2 \cos \sqrt{3x^2+4}}{e^{2x} \sin \sqrt{3x^2+4}}$$

$$= \frac{-3x - 2 \cos \sqrt{3x^2+4} (\sqrt{3x^2+4}) (\sin \sqrt{3x^2+4})}{e^{2x} \sqrt{3x^2+4} \cdot \sin^2 \sqrt{3x^2+4}} = \frac{-3x - \sin(2\sqrt{3x^2+4}) (\sqrt{3x^2+4})}{e^{2x} \sqrt{3x^2+4} \cdot \sin^2 \sqrt{3x^2+4}}$$

$$4) y = e^x \sinh(3-2x)$$

$$y = e^x, y' = e^x$$

$$v = \sinh 3-2x, v' = -2 \cos(3-2x)$$

$$\frac{dy}{dx} = e^x (-2 \cos 3-2x) + e^x (\sinh 3-2x)$$

$$= -2e^x \cos 3-2x + e^x \sinh 3-2x$$

$$a = -2e^x \cos 3-2x$$

$$v = e^x \sinh 3-2x$$

$$a = -2e^x$$

$$c = e^x$$

$$b = \cos(3-2x)$$

$$c' = e^x$$

$$a' = -2e^x$$

$$d = \sinh 3-2x$$

$$b' = 2 \sin(3-2x)$$

$$d' = -2 \cos(3-2x)$$

$$-3e^x \sinh 3-2x - 4e^x \cos 3-2x +$$

$$(4e^x \cos 3-2x) - (2e^x \sinh 3-2x) + 5y$$

$$-5e^x \sinh 3-2x + 5y = 0$$

$$-e^x \sinh 3-2x + \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = e^x \sinh 3-2x$$

$$y = e^x, y' = e^x$$

$$v = \sinh(3-2x)$$

$$v' = -2 \cos(3-2x)$$

$$= -2e^x \cos 3-2x + e^x \sinh(3-2x)$$

$$\frac{dy}{dx} = -4e^x \sinh 3-2x - 2e^x \cos 3-2x$$

$$\frac{dv}{dx} = -2e^x \cos 3-2x + e^x \sinh 3-2x$$

$$\frac{d^2y}{dx^2} = -3e^x \sinh 3-2x - 4e^x \cos 3-2x$$

$$5) y^3 = \ln(xy^6), x > 0, y > 0$$

$$3y^2 \frac{dy}{dx} = \frac{dy}{dx} (\ln(xy^6))$$

$$3y^2 \frac{dy}{dx} = \frac{dy}{dx} (\ln(xy^6))$$

$$3y^2 \frac{dy}{dx} = \frac{y^6 + 6xy^5 \frac{dy}{dx}}{xy^6}$$

$$3y^2 \frac{dy}{dx} = \frac{y^6 + 6xy^5 \frac{dy}{dx}}{xy^6}$$

$$3y^2 \frac{dy}{dx} = \frac{y + 6x \frac{dy}{dx}}{xy}$$

$$3xy^3 \frac{dy}{dx} = y + 6x \frac{dy}{dx}$$

$$3xy^3 \frac{dy}{dx} - 6x \frac{dy}{dx} = y$$

$$3(xy^3 - 2x) \frac{dy}{dx} = y$$

$$\frac{dy}{dx} = \frac{y}{3(xy^3 - 2x)}$$

$$y = \ln(xy^6)$$

$$y = \ln(u)$$

$$\frac{dy}{dx} = \frac{1}{u}$$

$$\frac{dy}{dx} = \frac{y^6 + 6xy^5 \frac{dy}{dx}}{xy^6}$$

$$\frac{dy}{dx} = \frac{1}{3e^1 - 6}$$

$$= 0.464$$

$$u = xy^6$$

$$\frac{du}{dx} = u' + u'v$$

$$= x(6y^5) \frac{dy}{dx} + y^6$$

$$= y^6 + 6xy^5 \frac{dy}{dx}$$

$$v = x, v' = 1$$

$$w = y^6, w' = 6y^5 \frac{dy}{dx}$$